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17 November 1977
SUPERSEDING
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MILITARY SPECIFICATION

DETECTOR KIT, WATER, AUTOMOTIVE AND AVIATION FUELS,
CAPSULE INDICATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of water detecting kit, suitable for detecting free water in automotive and aviation fuels.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified therein.

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Shipping, Fiberboard.

MILITARY

MIL-D-3464	-	Desiccants, Activated, Bagged Packaging Use and Static Dehumidification.
MIL-L-10547	-	Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MS 20003	-	Indicator, Humidity, Card Three Spot, Impregnated Areas (Cobaltous Chloride)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, and shall be suitable in all respects for the purpose intended.

* 3.2 Style and design. The water detecting kit shall consist of 625 capsules, 25 jars, 25 packages of desiccant, 25 indicator cards and 25 instruction sheets.

* 3.2.1 Capsules. The capsules shall be furnished 25 in each of the 25 jars. The capsules shall be made of a suitable clear plastic that will be inert to the substance it encloses, in two-piece construction, cylindrical in shape, with tapered walls, rounded ends, and a friction fit telescoping-type closure. Each capsule shall be 5/16 inch ± 1/8 inch in diameter and 7/8 inch ± 1/8 inch long and shall contain approximately 1/2 gram of anhydrous indicator powder.

* 3.2.1.1 Color change. In the presence of a minimum concentration of 30 parts per million (ppm) of water in 100 milliliters (ml) of fuel sample, the color of the powder shall change to pink or purple. If the concentration of water is less than 30 ppm, there shall be no change in color of the powder. The capsules shall be tested for color change in accordance with 4.4.

3.2.2 Jars. The kit shall contain 25 jars made of clear soda-lime glass. Jars shall be screw capped, round straight-walled, and shall have a capacity of 4 ounces. Each jar shall have a line 1 ± 1/16 inch long, permanently marked on the glass, at approximately the 100 ml ± 5 ml mark to indicate the volume of sample to be used. Each jar shall contain at least 1/2 unit of desiccant in accordance with MIL-D-3464. Packaging of desiccant shall be such that the atmosphere of the jar is exposed to the desiccant and the desiccant does not spill into the jar. In addition, each jar shall contain an indicator card, whose indications are easily observed from outside of the jar. The indicator card shall be in accordance with MS 20003. The jar shall be sealed with a plastic band, or other suitable means.

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3.3 Instructions. Each kit shall be furnished with a set of instructions describing the test procedure. The instructions may be furnished on the label of each bottle, or by an instruction card.

* 3.4 Marking. In addition to the marking requirements specified in 3.2.2 each jar shall be permanently marked with the following information: nature of contents, quantity, manufacturer's name or registered trademark, and the words "DANGER, Poisonous If Swallowed" and the listing of the compound if a poisonous substance is used in the manufacture of the indicator powder.

3.5 Workmanship. Workmanship shall be first class throughout. Kits shall be free from defects which impair their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the components, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of issue for examination purposes shall be one complete detector kit of 625 capsules in 25 jars.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II.

TABLE II. Sampling for tests.

Unit of product	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Kit	Color change	3.2.1	4.4	S-1	*

* Acceptance number shall be zero for all sample sizes.

4.3 Examination. The detector kits shall be examined to determine compliance with all of the requirements of this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III.

TABLE III. Classification of defects.

Categories	Defect
Major	
101	Jar lids not sealed properly
102	Glass jar cracked
103	Indicator card indicated exposure to excessive humidity, or missing from jar
104	Capsule contents not uniform in color
105	Capsule contents or desiccant contents loose in jar
Minor	
201	Instructions missing
202	Marking missing or illegible (see 3.4)

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4.4 Tests.

* 4.4.1 Stock preparation. A test sample shall be prepared as follows. Pour one liter of bright, clear, kerosene type fuel into a clean container which shall be capped at all times between operations. Saturate a blotter or heavy filter paper with a minimum of 10 ml of water. Place the saturated blotter or filter paper at the bottom of the container of kerosene. Allow the container to set, without agitation, for a period of 24 hours to insure saturation.

4.4.2 Sample preparation. Five samples shall be prepared. Add 100 ml of the stock preparation (4.4.1) to each of five, four ounce, clean, dry, glass bottles. Using a micropipet, microburet or microsyringe add the following measured amounts of distilled water to the sample bottles as indicated below:

Sample	Distilled water to be added	Suspension (free water)
1	0	0 ppm
2	1 microliter	10 ppm
3	2 microliter	20 ppm
4	3 microliter	30 ppm
5	4 microliter	40 ppm

Cap the bottles and shake vigorously for one minute using a mechanical shaker to obtain a fine dispersion of the water. Avoid a change of temperature. Uncap the bottles and add to each the contents of one capsule from the kit. Recap the bottles and shake vigorously, by hand, for a minimum of 10 seconds. Examine the powder for any evidence of color change in accordance with 4.4.3. Ignore any color change that occurs after 2 minutes.

4.4.3 Color change. At the end of the ten seconds the powder in the five bottles shall be examined for color change in accordance with 3.2.1. In the first bottle to which no water was added and in the second bottle to which 10 ppm of free water was added, no color change should occur. In the third bottle to which 20 ppm of free water was added, either no color change or only a faint color change should occur. In the fourth bottle to which 30 ppm of free water was added, a significant color change shall be seen, and in the fifth bottle to which 40 ppm of free water was added, a pronounced color change shall be apparent. When these color changes occur as specified, the detector kit is acceptable for use.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

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TABLE IV. Examination for preparation for delivery.

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages or shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PACKAGING

5.1 Preservation - packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. Each kit shall be packaged in a box of an appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. The box shall include sufficient cushioning material to prevent damage to the contents. Closure shall conform to method I or II as specified in the appendix of PPP-B-636.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Each kit packed as specified in 5.1.1 shall be overpacked in an exterior container designed for a type I load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

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5.2.2 Level B. Kits packaged as specified in 5.1.1 shall require no further overpacking.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure accpetance and safe delivery called for in the contract or purchase order.

5.3 Marking. Marking shall conform to MIL-STD-129. Marking shall include the legend "GLASS - HANDLE WITH CARE" and the word "POISON" as required by the applicable statues regulations.

6. NOTES

6.1 Intended use. The detector kits covered by this specification are intended to be used in the detection of free water in automotive or aviation fuels.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN) of this item.
c. Selection of applicable levels of packaging and packing. (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-892-2264	Detector Kit, Water, Automotive and Aviation Fuels

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6640-1019

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-D-36342B DETECTOR KIT, WATER, AUTOMOTIVE AND AVIATION FUELS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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